

OIA v0.5 — Architecture Lock

Specification of the v0.5 architectural changes ahead of the document family upgrade cycle. Reference document for the Reference Paper, the Digest, the Lens System, the architecture diagrams, and the thought leadership deck.

Internal working document. Not for external distribution.

Version 0.5 · Lock candidate · 2026

1. Purpose of This Document

The OIA Reference Model is being upgraded from v0.1 to v0.5 to incorporate the architectural refinements that emerged from the Proofpoint pressure test, and to renumber the layer count for mathematical and rhetorical honesty. The upgrade affects the full document family — the Reference Paper, the Digest, the Lens System, the architecture diagrams now in active development, and the thought leadership deck. To prevent drift across these artifacts, the v0.5 architecture must be locked canonically before the heavy content work begins.

This document is the lock. It specifies, in compressed form, what changes from v0.1 to v0.5: the ten-layer renumbering, the seven architectural refinements, the new cross-layer span list, and the canonical vocabulary that all family documents must use consistently. It also specifies what does not change — because the upgrade is additive and the architectural spine of v0.1 is preserved.

This is a working document. It is not intended for external distribution. Its purpose is to give every family deliverable a single reference against which architectural language can be checked. Once the family is published at v0.5, this document is retired into the Foundation's working archive.

2. Ten-Layer Renumbering

2.1 What changes

The OIA Reference Model is described henceforth as a ten-layer model, numbered 0 through 9. Every reference to "nine layers" in v0.1 documents is replaced with "ten layers, numbered 0 through 9" or contextual variants of the same. Layer 0 retains its name (Physical Compute) and its place at the foundation of the stack. Nothing about the architecture itself changes; only the count is corrected.

2.2 Why

The v0.1 description of the model as nine-layer was a holdover from OSI parallelism that did not match the architecture as drawn. Layers 0 through 9 inclusive is ten layers. The internal inconsistency was a credibility issue — most acutely in the v0.1 Foreword, which described "nine layers, from physical compute at the bottom," a phrasing that is technically self-contradicting because physical compute is Layer 0.

Calling it ten layers preserves Layer 0's distinct status (the physical substrate beneath the architectural stack) while being mathematically honest about the count. The OSI bottom-up

numbering convention is preserved. The departure from OSI — introducing Layer 0 — remains principled and explained.

2.3 Canonical phrasings

Where v0.1 used "nine layers," v0.5 uses one of the following canonical phrasings depending on context:

"Ten layers, numbered 0 through 9, with Layer 0 the physical substrate beneath the architectural stack."

"Ten layers — from physical compute at Layer 0 to the human and browser interface at Layer 9."

"The OIA's ten layers and seven cross-layer spans."

The phrasing "the architectural stack" is preferred when emphasising Layers 1 through 9 as a logical group, with Layer 0 understood as the substrate that grounds the stack. This phrasing is helpful in audience-facing materials where the categorical distinction of Layer 0 (physical, sub-architectural) versus Layers 1–9 (logical, architectural) is useful.

3. The Seven Refinements

Seven architectural refinements emerge from the Proofpoint pressure test and are absorbed into v0.5. Each is defined here in canonical form. None of them changes the layer count or the layer names; all of them sharpen the architecture against what the lens work surfaced. Family documents must use the v0.5 definitions verbatim or in faithful paraphrase.

3.1 Verifiability — a seventh cross-layer span

v0.5 elevates verifiability from an implicit property of Layer 1 and the security span to an explicit cross-layer span in its own right. The new Verifiability span is defined as the property that operations of an intelligent system can be cryptographically proven rather than contractually asserted. It threads through Layers 1, 2, 4, and 5 in particular, anchored by hardware-attested execution (Intel TDX, AMD SEV-SNP, NVIDIA Confidential GPU) and the Confidential Computing Consortium body of work, and extending to witness-chain logic at Layer 8.

Canonical definition. The Verifiability span captures the property that the operation of an intelligent system can be cryptographically proven, not contractually asserted. Trust is verified rather than granted. Threads primarily through Layers 1, 2, 4, 5, and 8.

3.2 Three actors — strengthening the Identity span

v0.5 strengthens the Identity span by naming three actors explicitly: humans, agents, and foundation models. The third actor — model-as-actor with a continuously refreshable trust score — is new architectural recognition. Models are no longer treated as static dependencies; provider weight swaps, alignment drift, jailbreak emergence, and LLM-specific CVEs make model trustworthiness dynamic and warrant continuous trust governance alongside human and agent identity.

Canonical definition. The Identity span addresses three classes of actor inside the enterprise: humans, agents, and foundation models. Each class carries a distinct identity lifecycle and trust posture. Models are actors whose trustworthiness is dynamic and must be continuously scored against current threat reality.

3.3 Layer 8 reframed to include recovery and rollback

v0.5 reframes Layer 8 (Continuity Fabric) to include recovery and rollback explicitly alongside the existing continuity-of-cognitive-state framing. The reframing is additive — the abstract framing remains accurate — but the operational language gives Layer 8 a concrete failure mode and a concrete responsibility that practitioners across audiences can engage with.

Canonical definition. Layer 8 — Continuity Fabric. The active layer at which cognitive state, learning, and judgment are preserved, verified, and carried across the transitions the stack undergoes; and at which recovery, rollback, and post-incident integrity for autonomous systems are made architecturally tractable.

3.4 Agent-in-the-middle named at Layer 7

v0.5 adds agent-in-the-middle to the Key Concerns of Layer 7 (Orchestration and Workflow). The agent-in-the-middle pattern — a rogue MCP server inserting itself between a trusted agent and a real tool, relaying sanitised responses while exfiltrating — is a recognised threat at the orchestration layer that v0.1 did not name explicitly.

Canonical addition. Layer 7 Key Concerns now include agent-in-the-middle attacks (the agentic SSL strip), where adversarial intermediaries operate transparently between an agent and a trusted tool surface, alongside the existing concerns of intent verification, action auditability, tool integrity, and kill-chain detection.

3.5 AI-Inside named at Layer 3

v0.5 adds the AI-Inside concern to Layer 3 (Agent Data Substrate). The structural observation that essentially every modern SaaS now embeds LLMs creates a silent data flow at Layer 3 that traditional DLP and data loss prevention approaches do not see. Naming it makes the concern architecturally visible.

Canonical addition. Layer 3 Key Concerns now include the AI-Inside problem: embedded-LLM data flow visibility. Most enterprise SaaS now embed foundation models invisibly, creating data egress paths to model providers that traditional DLP frameworks were not designed to detect or govern.

3.6 Temporal validity — absorbed into Consequence

v0.5 acknowledges that the correctness of an intelligent system's posture is time-dependent in a way infrastructure correctness is not. A cleared CVE, a patched library, a swapped model, a freshly emerged jailbreak — each of these changes correctness mid-flight. Rather than introducing a fourth defining property of intelligent systems, v0.5 absorbs this into the existing third property (Consequence) by adding a temporal dimension.

Canonical extension. Consequence (the third defining property of intelligent systems): intelligent systems produce outputs and take actions whose correctness, accountability, and auditability must be demonstrable — and demonstrable against current threat reality, not against the threat reality at the moment of deployment. Architectural posture is time-bound.

3.7 The four named threat surfaces

v0.5 incorporates the four named threat surfaces of the agentic era — mimicry, agent-in-the-middle, agent spoofing, model compromise — into the model's threat vocabulary. These threats are not introduced as a new architectural element; they are named in the appropriate layer Key Concerns and in the threat-surface diagram. The naming gives security audiences a shared vocabulary that ties to the architecture.

Canonical naming. The four named threat surfaces of the agentic era are: Mimicry (Layer 9 propagating through Layers 7 and 3); Agent-in-the-middle (Layer 7); Agent spoofing (Identity span, Layers 5 through 9); and Model compromise (Layer 5, Verifiability and Security spans).

4. Updated Cross-Layer Spans

v0.5 raises the cross-layer span count from six to seven. The six v0.1 spans are preserved unchanged in scope. Verifiability is added as the seventh, capturing the architectural property surfaced during the Proofpoint pressure test. The full v0.5 cross-layer span list is canonical:

- **Security.** The cross-layer concern with the widest scope. Every layer has security properties; coherence requires guarantees at lower layers to surface to higher layers without being lost.
- **Sovereignty.** Three dimensions — jurisdictional, operational, energy — established at Layer 2 and threading through every layer above. Sovereignty posture is only as strong as the weakest layer.
- **Auditability.** The capacity to reconstruct, after the fact, what happened and why. Cannot be retrofitted; must be designed into every layer from the outset.
- **Identity.** Who is acting in any given operation — across three classes of actor: humans, agents, and foundation models. Foundational to auditability.
- **Energy and Environmental Concerns.** The energy consumed at Layer 0 propagates as cost, environmental impact, and capacity constraint to every layer above. Increasingly architectural rather than operational.
- **Provenance.** The lineage and origin of information, models, decisions, and actions. Threads from Layer 3 to Layer 9. Distinct from auditability: auditability is the capacity to reconstruct; provenance is the property that enables reconstruction.
- **Verifiability.** New in v0.5. The property that operations of an intelligent system can be cryptographically proven rather than contractually asserted. Trust verified, not granted. Threads primarily through Layers 1, 2, 4, 5, and 8.

The v0.5 phrasing "the OIA's ten layers and seven cross-layer spans" is canonical. v0.1 phrasings referring to "six cross-layer spans" are replaced. Documents that need to acknowledge the change explicitly may state that v0.5 introduces Verifiability as the seventh span, surfaced during the Proofpoint pressure test of the model.

5. Canonical Vocabulary

Family documents must use consistent terminology across the upgrade. The phrasings below are the canonical references. Faithful paraphrase is acceptable; substitution with synonyms that drift from the architecture is not.

5.1 Layer references

- **The architecture:** "the OIA's ten layers and seven cross-layer spans"
- **Layer naming:** Layer 0 (Physical Compute), Layer 1 (Silicon Abstraction), Layer 2 (Sovereign Infrastructure), Layer 3 (Agent Data Substrate), Layer 4 (Model Training and Adaptation), Layer 5 (Inference and Retrieval), Layer 6 (Context and Knowledge), Layer 7 (Orchestration and Workflow), Layer 8 (Continuity Fabric), Layer 9 (Human and Browser Interface).
- **Architectural stack vs. substrate:** Layers 1–9 are "the architectural stack." Layer 0 is "the physical substrate beneath the architectural stack."

5.2 The defining properties of intelligent systems

Three properties (unchanged from v0.1, with the temporal dimension absorbed into Consequence as specified in 3.6):

- **Persistence.** Intelligent systems maintain state across interactions. Addressed primarily at Layer 8, secondarily at Layer 3.
- **Autonomy.** Intelligent systems act with degrees of autonomy that exceed traditional application behaviour. Addressed primarily at Layer 7.
- **Consequence.** Intelligent systems produce outputs and take actions whose correctness, accountability, and auditability must be demonstrable, and demonstrable against current threat reality. Addressed through the cross-layer spans.

5.3 The four named threat surfaces

- **Mimicry.** The boundary problem of telling humans from agents at every entry point. Layer 9 origin, propagating through Layer 7 to Layer 3.
- **Agent-in-the-middle.** Adversarial intermediaries between agents and tools, including rogue MCP servers. Layer 7.
- **Agent spoofing.** Agent identities inheriting trust they should not have, across contexts and lifetimes. Identity span, Layers 5 through 9.
- **Model compromise.** The trusted model that is no longer trustworthy — weight swap, alignment drift, jailbreak emergence, LLM CVEs. Layer 5, Security and Verifiability spans.

5.4 The three actors

v0.5 names three classes of actor inside the enterprise that require continuous trust governance: humans, agents, and foundation models. The phrasing "three actors that require continuous trust governance — humans, agents, and the foundation models themselves" is canonical.

6. What v0.5 Preserves Unchanged

The upgrade is additive. The architectural spine of v0.1 is preserved without modification. Specifically:

- **All ten layer names are preserved.** The renumbering describes the existing layers; it does not relabel them.
- **All ten layer purposes are preserved.** The primary purpose statement of each layer is unchanged from v0.1.
- **The bottom-up numbering convention is preserved.** Layer 0 remains the foundation; Layer 9 remains the top.
- **The two paired state-holding layers are preserved.** Layer 3 (Agent Data Substrate) and Layer 8 (Continuity Fabric) remain the two state-holding layers, with Layer 3 a substrate and Layer 8 a fabric.
- **The relationship to other reference models is preserved.** The OIA continues to be additive to OSI, NIST CSF, NIST AI RMF, MITRE ATLAS, OWASP LLM, and ISO/IEC 42001 — not competitive with them.
- **The Open Questions structure is preserved.** Each layer continues to name what is unresolved. v0.5 closes none of the v0.1 open questions.

Reviewers familiar with v0.1 should find v0.5 immediately recognisable. The architecture is the same architecture, sharper at seven specific points, with the layer count corrected.

Closing

With the architecture locked at v0.5, the family upgrade can proceed. Architectural diagrams and the Reference Paper are the immediate next deliverables, working in parallel against this specification. The Digest follows the Reference Paper, the Lens System aligns at v0.3 to absorb v0.5 references, and the cover document and thought leadership deck are authored once the family is coherent.

Any drift from this specification observed during the upgrade cycle should be raised immediately and resolved either by correcting the deliverable or amending this specification. There is no stable equilibrium in which the deliverables and the lock disagree.